AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

# Genesis Ideathon: Supercapacitor-Integrated Hydraulic and Electromagetic retaders [S-IHER] Final Presentation

Team ID: *GEN26-A0H-I149* Team Name: *F.I.A* Domain: *Energy, Mobility and Smart Infrastructure*

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# Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
7. Feasibility & Scalability
8. Cost for Product
9. Use Cases
10. Impacts and Benefits
11. Sustainable Development Goals
12. Individual Member Contribution

# Problem Statement

When a heavy electric truck or a bus is fully charged or running too hot or cold, the system automatically shuts off regenerative braking to protect the battery. This forces the vehicle to rely entirely on mechanical brakes (Hydraulic brakes), causing extreme overheating, rapid pad wear, and dangerous brake fade on highways and slopes.

# Problem Understanding

- Heavy Electric Truck and Bus Drivers
- When the battery is fully charged or is at an extreme temperature which disables regen-braking, causing us to rely entirely on standard mechanical brakes to stop a massive, speeding vehicle. On long hills, these friction brakes get pushed past their limits. They quickly overheat, wear out, and suffer from dangerous brake fade—meaning they lose stopping power entirely.
- It affects the Fleet Owners and Logistics Companies who invest in electric trucks for emission goals but face high maintenance costs, brake degradation, and liability. It also affects OEMs and EV Engineers who are responsible for vehicle safety, battery health management, and regulatory compliance.

# Existing Gaps & Challenges

- Solutions to this problem already exist but they rely too heavily on mechanical breaking which can lead to break failure
- 1"Resistive" Regen-Braking Systems:- It converts the regen-electrical energy to heat by resistors. Key challenge: How can excess braking energy be stored or reused instead of being dissipated as heat.
- 2.Advanced Air-Braking Systems:- Existing solution: Electronic Brakes coordinate regenerative braking with pneumatic friction brakes to maintain controlled deceleration. Key challenge: How can braking safety be maintained while reducing friction-brake overheating and wear during long downhill descents.
- 3. Electromagnetic Braking:- Existing solution: uses magnetic field and eddy currents to induce retardation. Key challange: too ineffective in slow speeds so it has to rely on mechanical braking at some point.
- The main problem is to find a way to utilize the additional energy generated in a useful way, which at the same time should be a long term, reliable as well as a cost-effective solution to this problem.

# Our Proposed Solution

- Our core idea is to develop a system that uses regenerative braking to the maximum by utilizing the effectiveness of electromagnetic braking at high speeds, excess energy storage with the help of super capacitors while keeping hydraulic braking as a backup.
- Our system works by capturing braking energy, storing it in the battery and supercapacitor, and using the supercapacitor as a fast boost source to strengthen the electromagenetic retarder's braking output, with normal brakes handling the final stop.
- The main value we deliver to users through this solution is more reliable and long lasting braking units for extremely heavy vehicles.

# Technical Approach

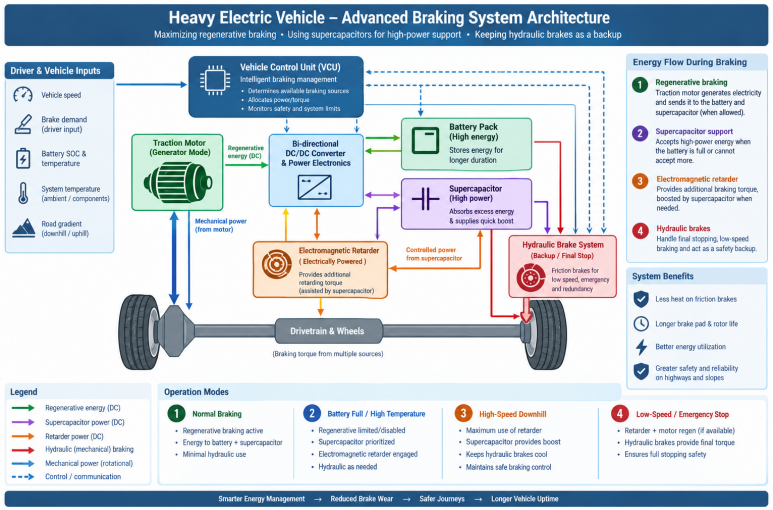



**Figure:** .

architectural flowchart

**Tech Stack**

**Frontend:** React dashboard

**Backend:** Python API service

**Database:** PostgreSQL + time-series sensor store

**AI/ML, APIs, or Hardware:** MPC + LSTM/CNN models + XAI or CAN bus + edge controller

# Innovation & Uniqueness

- The genuinely new part is not the individual components, but the way you combine them into one braking architecture: an IEHR that still works independently, plus a supercapacitor that is used as an active torque-support layer, not just as energy storage for regenerative braking. but the main part is the system that we have to develop to manage the braking force output.

- This approach is suited to the problem because it gives us a efficient way of using the energy generated in regen-braking and also provides a safer alternative than existing systems.

- Our idea is best positioned as a hybrid braking architecture that adds supercapacitor-assisted torque boosting to an IEHR, rather than simply another regenerative braking or retarder product. In practical terms, the competitive difference is not a new brake physics principle, but how the braking, storage, and control layers are combined to deliver more usable braking support.

# Feasibility & Scalability

- **Feasibility:** Vehicle  Powertrain Components: Electric motors, power electronics (inverters/converters), and auxiliary energy storage/dissipation hardware (e.g., supercapacitors, advanced thermal management systems, or modular auxiliary packs). Sensors  Control Units: High-precision battery management system (BMS) sensors, temperature sensors,etc.. Testing Prototyping Gear: Dynamometers, thermal imaging cameras (to monitor friction brake and resistor temperatures), and hardware-in-the-loop (HIL) test benches.

- **Scalability:** Growing it by proving one high-value use case, standardizing it as a modular retrofit/product, and selling the control + maintenance layer as much as the hardware.

- **Risks:** The main constraint is that we need to develop a high level system that can monitor the speed, incline, and other factors to choose suitable constraints to provide adequate amount of breaking.

# Cost for Product (Optional)

- Development or hardware cost: 7000 euros approx
- Hosting or maintenance cost: 100 euros approx
- Estimated total: 7100 euros approx

# **Use Cases** (compulsorily 3 use cases must be there)

- Use case 1: Bus or Truck driver driving heavy trucks and buses downhill
- Use case 2: An urban bus or delivery vehicle driver brakes repeatedly at stops, signals, and low-speed traffic in the city.
- Use case 3: Rail operators can use these in Diesel-hauled commuter trains or heavy-haul rail lines recover braking energy on long routes

# Impacts and Benefits

- Truck Drivers gain a safe, reliable, and predictable braking system that doesn't put their lives at risk when starting a route with a 100 percent charged battery. They save significant money on brake replacement and maintenance, avoid costly accident liabilities, and no longer lose cargo capacity to heavy, dead-weight resistor grids. Highway Motorists and the general public are protected from the catastrophic hazard of runaway 80,000-pound trucks careening down mountain passes.
- The biggest positive impact of your idea is in commercial and public fleets that brake often, where it can simultaneously improve safety, lower brake wear, save energy, and cut emissions.
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# Sustainable Development Goals (SDGs)

| SDG Number | Target Number | Goal Name | Connection |
|---|---|---|---|
| **SDG 7** | 7.1, 7.2, 7.3 | Energy access, renewable energy adoption, and energy efficiency | by utilising various braking metods we maximize efficiency and generate energy in the process. |
| **SDG 9** | 9 (overall) | Industry inovation and infrastructure | By using the regen-energy effectively for braking, we also minimised the pollution caused by the brake pads and disk. |

# Individual Member Contribution

| Member | Contribution or responsibility |
|---|---|
| **V Vishvanth Ram** | [Contribution or responsibility] |
| **Vedant Vishwas Chopade** | [Contribution or responsibility] |
| **Gollamudi Srirama Kanvas** | [Contribution or responsibility] |
| **Topalle Venkata Ananta Sai Subrahmanyam** | [Contribution or responsibility] |

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